

Reformulation for Formula E Style Energy-Aware RL in TORCS

1. Why we want to pivot the framing

In the earlier version, we framed the project as generic fuel-aware racing. After feedback, we think that framing is too weak from a real-world impact perspective because it sounds like a simulator-only racing exercise.

We now want to reframe the project around:

Formula E style energy management for autonomous racing

The motivation we want to emphasize is not simply "saving fuel in TORCS." What we actually want to study is:

- how to manage a limited energy budget over a race,
- how to decide when to drive aggressively and when to conserve,
- and how to learn a controller that adapts online to the remaining usable energy.

We think this connects much more clearly to:

- Formula E style electric racing,
- hybrid race car energy strategy,
- autonomous electric racecars,
- and, more broadly, battery-constrained robotic control.

2. What the current project already supports

Based on the work already done in this repo:

- native fuel / tank / RPM telemetry is available on the RL side,
- the environment can track per-step energy usage,
- the episode can terminate on energy depletion,
- and the codebase already supports a baseline single-car racing workflow.

So we do not think we need to invent a completely new project. We think we already have the infrastructure needed for an energy-management problem. What we needed to improve was the problem formulation and the real-world motivation.

3. Proposed reformulated problem

Budget-Conditioned Energy Management for Time-Efficient Racing

We want to study a policy of the form:

$\pi(a_t \mid o_t, b_t)$

where:

- o_t is the current observation,
- b_t is the remaining energy budget,
- and the policy chooses actions that balance pace and conservation.

The objective we want is:

- complete the lap or race segment as quickly as possible,
- do not exceed the available energy budget,
- and adapt the driving style online as the remaining budget changes.

Plain-language version

We want to learn one driving policy that can behave like an aggressive attack mode when energy is plentiful and a conservation mode when energy is tight, without training a separate policy for every budget.

4. Why we think this has real-world impact

We think this formulation has a much stronger real-world story if we present it as a Formula E style control problem.

Direct impact

- **Electric race strategy:** In electric racing, the car cannot just maximize instantaneous speed. It has to allocate limited usable energy over the race.
- **Autonomous electric racecars:** If the vehicle is autonomous, the controller itself has to make local pace-versus-energy decisions in real time.
- **Hybrid / energy recovery systems:** More generally, deciding when to spend and when to conserve stored energy is already a real motorsport optimization problem.

Indirect impact

- **EV eco-driving:** The same class of method is relevant for battery-aware driving policies that trade trip time against energy use.

- **Robotics:** Any autonomous system operating under a hard battery constraint faces the same structure: limited energy, uncertain future demand, and online control decisions.
- **Runtime adaptation:** A single conditioned policy is practically useful when the available energy budget changes during deployment and solving a fresh optimization problem every time is expensive or inconvenient.

5. Why we think this is better than the previous framing

Previously, the project was closer to the question:

- can an RL policy trade lap time against fuel?

Now, we want the project to answer:

- can a single policy learn Formula E style energy management behavior and adapt online to remaining energy?

We think this is better because:

- it gives the work a clearer real-world analogue,
- it aligns the project with electric racing instead of generic simulator fuel penalties,
- and it makes the budget-conditioned policy idea feel like a deployment-oriented controller rather than only a benchmark trick.

6. What we want to claim as the contribution

We do not want to claim something too broad like "RL for energy-aware racing," because that would be weak.

Instead, we want the contribution to be:

1. **A budget-conditioned energy-aware racing policy**
We want to learn a single policy that adapts continuously to the remaining energy budget.
2. **Runtime adaptation instead of one-policy-per-budget training**
We want the controller to change behavior online without retraining for each energy target.
3. **Explicit resource state in the policy input**
We want remaining energy to be part of the decision process, not just hidden inside a fixed penalty term.
4. **Evaluation across multiple and unseen budgets**
We want to test whether the learned policy generalizes across different energy allocations.

7. Research question and hypothesis

Main question

We want to answer the following question:

Can a single energy-budget-conditioned RL policy learn Formula E style pace-versus-conservation behavior and outperform:

- an unconstrained lap-time policy,
- a fixed energy-penalty policy,
- and separate one-policy-per-budget baselines?

Hypothesis

Our hypothesis is that a budget-conditioned policy will:

- recover near-best lap performance when the budget is loose,
- avoid budget exhaustion better than a fixed-penalty baseline when the budget is tight,
- and produce a smoother energy-versus-time tradeoff across the full budget range.

8. How we want to frame the experiments

To keep the project feasible, we still want to use TORCS as the experimental environment. We think the important point is to describe that setup carefully:

- TORCS is the benchmark environment,
- the energy budget is a proxy for usable race energy,
- and the intended application is Formula E style energy management, not literal direct deployment from TORCS.

We think this is a reasonable research framing:

- simulator as a controlled testbed,
- real-world application as energy-aware electric racing control.

9. What we want to compare against

Baselines

1. **Unconstrained pace policy**
A policy that maximizes speed or progress with no explicit energy awareness.
2. **Penalty-shaped energy policy**

A standard reward-shaping approach with a fixed energy penalty.

3. **Separate policy for each budget**

A stronger baseline, but one that is operationally inconvenient.

Proposed method

4. **Single budget-conditioned policy**

Our main method, where the policy receives normalized remaining energy and adapts online.

10. How we want to evaluate it

The main metrics we want to use are:

- lap completion rate
- lap time or normalized progress
- total energy used
- energy budget violation rate
- off-track / terminal failure rate
- performance on held-out budgets
- time achieved per unit energy

If time allows later, we would also like to test:

- transfer to a second track
- robustness under different initial budgets

11. Sources supporting this framing

Formula E / race energy management

- Limebeer, Perantoni, Rao, *Optimal control of Formula One car energy recovery systems*, International Journal of Control, 2014.
<https://doi.org/10.1080/00207179.2014.900705>
- Applied Energy work on Formula E energy management under regulatory and thermal constraints.
<https://www.sciencedirect.com/science/article/pii/S0306261920312861>
- Applied Energy work on hybrid electric race car energy management and time-optimal strategy.
<https://www.sciencedirect.com/science/article/pii/S0306261920314288>
- Herrmann et al., *Energy Management Strategy for an Autonomous Electric Racecar using Optimal Control*.
<https://portal.fis.tum.de/en/publications/energy-management-strategy-for-an-autonomous-electric-racecar-usi/>

Budget-conditioned / constrained RL support

- Carrara et al., *Budgeted Reinforcement Learning in Continuous State Space*, NeurIPS 2019.
<https://research.google/pubs/budgeted-reinforcement-learning-in-continuous-state-space/>
- Achiam et al., *Constrained Policy Optimization*, 2017.
<https://arxiv.org/abs/1705.10528>

Broader energy-aware driving support

- Yang et al., *Eco-driving strategies using reinforcement learning for mixed traffic in the vicinity of signalized intersections*, Transportation Research Part C, 2024.
<https://www.sciencedirect.com/science/article/pii/S0968090X24002043>
- Zhang et al., *Navigating Electric Vehicles Along a Signalized Corridor via Reinforcement Learning*, Transportation Research Record, 2022.
<https://journals.sagepub.com/doi/10.1177/03611981221084683>

12. Final position

At this stage, we do not think we should pivot away from the method itself. We think we should pivot the application framing.

The title we currently find most appropriate is:

Learning a Formula E Style Energy-Management Policy with Budget-Conditioned RL

We think this gives the project:

- a stronger real-world motivation,
- a more defensible impact story,
- and a clearer explanation for why a single adaptive policy matters.